

INDEX

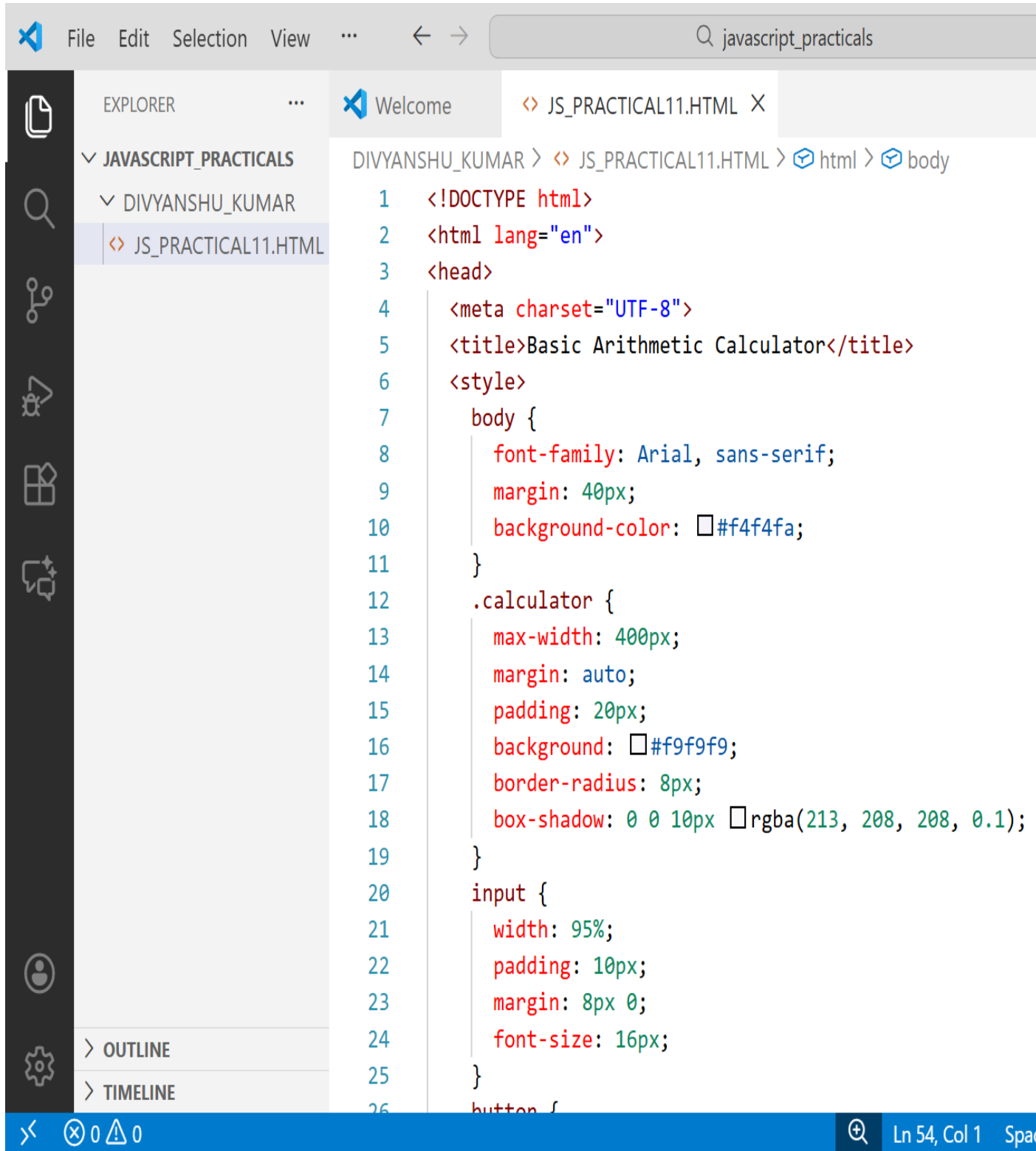
Sr. No	NAME OF Experiments	Submission date	Sign	Remarks
11	Create a JavaScript-enabled web page that performs basic arithmetic operations using input from users			
12	Write JavaScript to demonstrate conditional statements and looping (for, while, do...while)			
13	Create and invoke user-defined functions in JavaScript; use var, let, and const for scope demonstration			
14	Handle HTML form validation using JavaScript events like onsubmit, oninput, and use alert() for feedback			
15	Demonstrate DOM manipulation using getElementById(), querySelector(), and innerHTML			
16	Show or hide HTML elements using JavaScript and toggle CSS classes dynamically			
17	Add interactivity using event listeners (addEventListener, removeEventListener) for mouse/keyboard events			
18	Create and manipulate elements using JavaScript: append, remove, or modify child nodes			
19	Create a simple drawing using the element: draw shapes and fill colors			
20	Initialize a Git repository, commit changes, and push to GitHub; Deploy a personal portfolio site on GitHub Pages or Netlify			

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EXPERIMENT -11

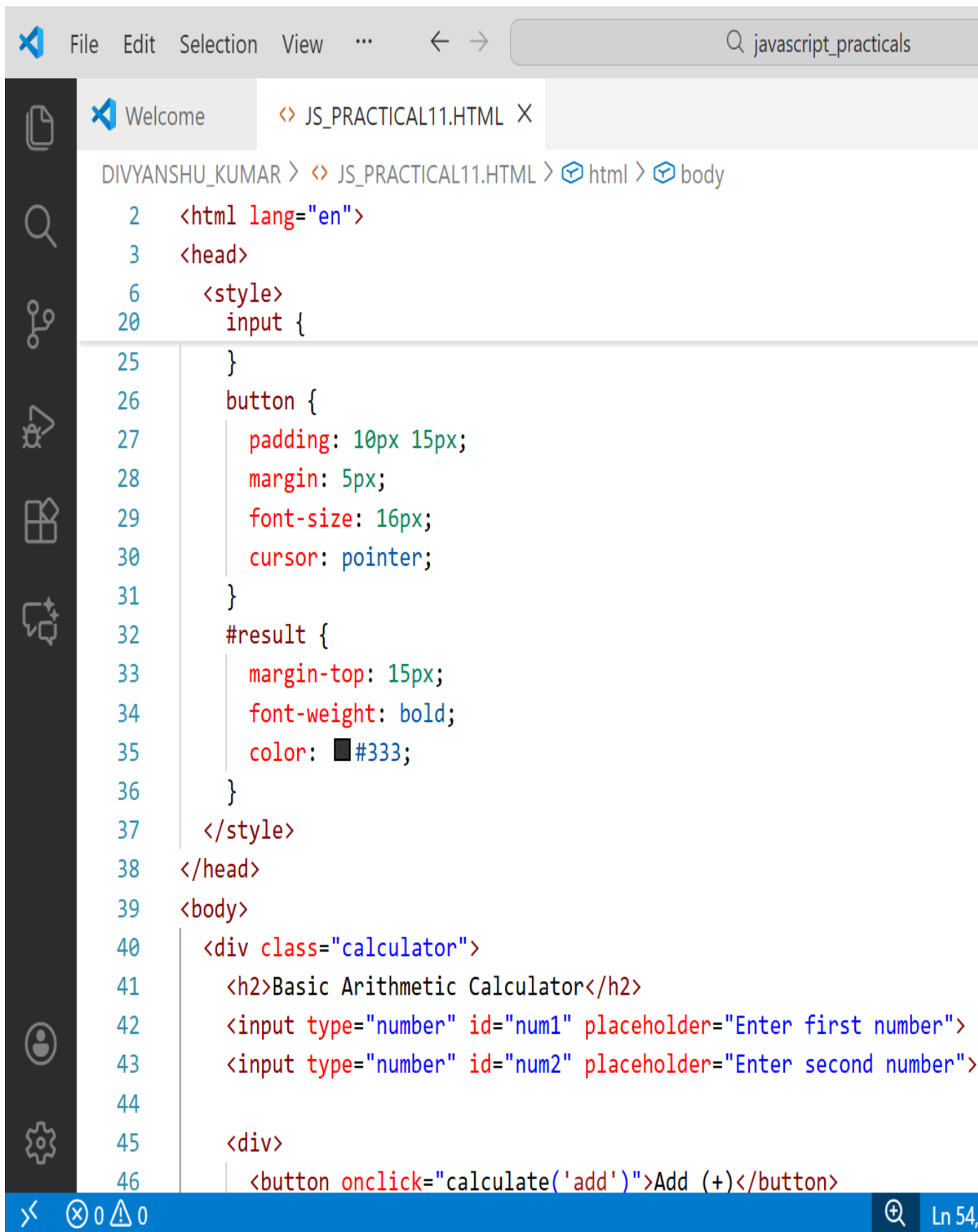
Problem statement: Create a JavaScript-enabled web page that performs basic arithmetic operations using input from users

HTML+CSS+JS CODE:



The screenshot shows the Visual Studio Code editor interface. The Explorer panel on the left shows the project structure with 'JAVASCRIPT_PRACTICALS' and 'DIVYANSHU_KUMAR' folders, and 'JS_PRACTICAL11.HTML' file selected. The main editor area displays the code for 'JS_PRACTICAL11.HTML'. The code includes a DOCTYPE declaration, HTML lang attribute, and a head section with a meta charset, title, and a style block. The style block defines the body and a calculator container with various CSS properties like font-family, margin, background-color, max-width, padding, background, border-radius, box-shadow, width, padding, margin, and font-size. The code is as follows:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Basic Arithmetic Calculator</title>
6   <style>
7     body {
8       font-family: Arial, sans-serif;
9       margin: 40px;
10      background-color: #f4f4fa;
11    }
12    .calculator {
13      max-width: 400px;
14      margin: auto;
15      padding: 20px;
16      background: #f9f9f9;
17      border-radius: 8px;
18      box-shadow: 0 0 10px rgba(213, 208, 208, 0.1);
19    }
20    input {
21      width: 95%;
22      padding: 10px;
23      margin: 8px 0;
24      font-size: 16px;
25    }
26    button {
```

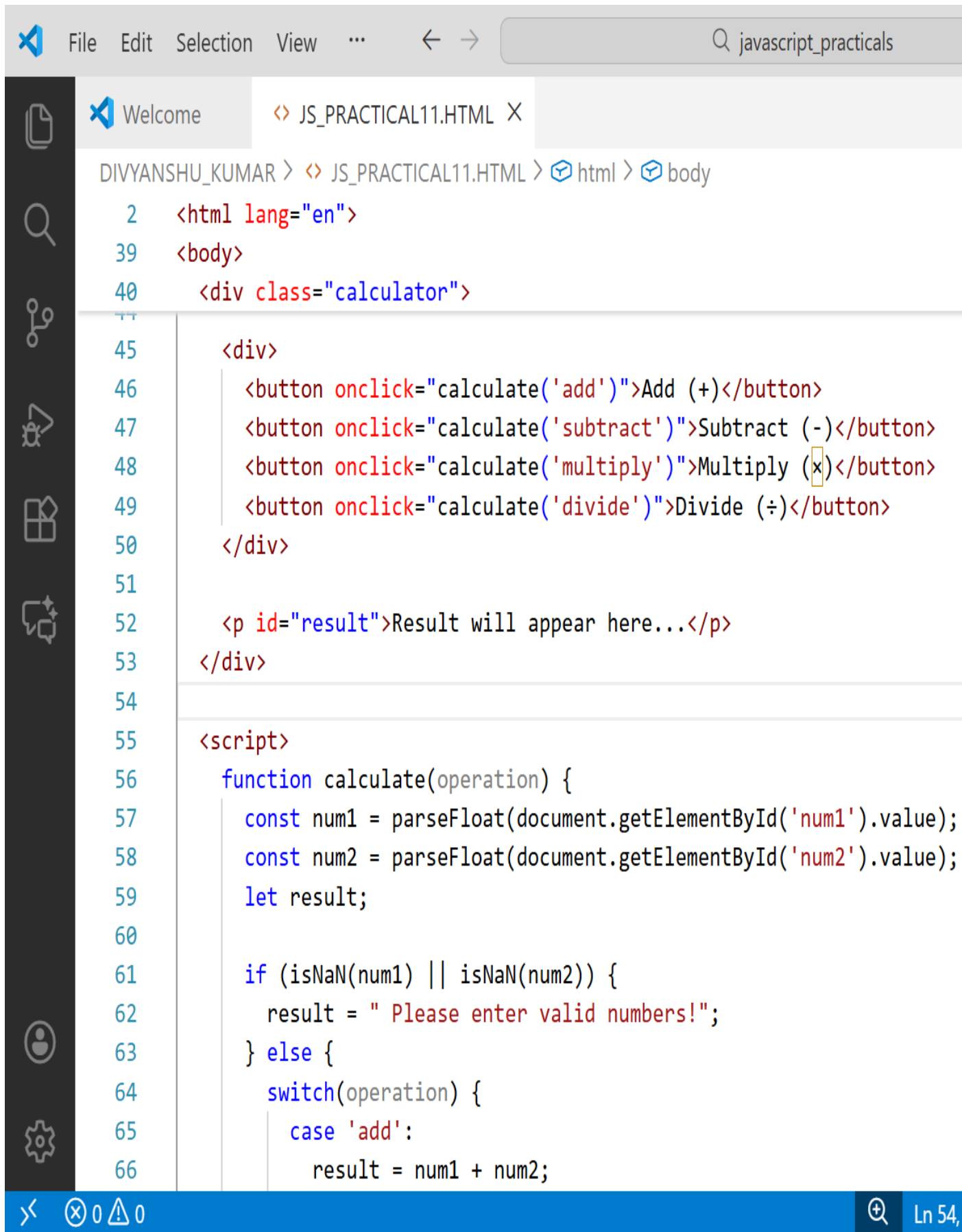


```
File Edit Selection View ... < > javascript_practicals

Welcome JS_PRACTICAL11.HTML X

DIVYANSHU_KUMAR > JS_PRACTICAL11.HTML > html > body

2  <html lang="en">
3  <head>
6    <style>
20    input {
25    }
26    button {
27      padding: 10px 15px;
28      margin: 5px;
29      font-size: 16px;
30      cursor: pointer;
31    }
32    #result {
33      margin-top: 15px;
34      font-weight: bold;
35      color: #333;
36    }
37  </style>
38 </head>
39 <body>
40   <div class="calculator">
41     <h2>Basic Arithmetic Calculator</h2>
42     <input type="number" id="num1" placeholder="Enter first number">
43     <input type="number" id="num2" placeholder="Enter second number">
44
45     <div>
46       <button onclick="calculate('add')">Add (+)</button>
```



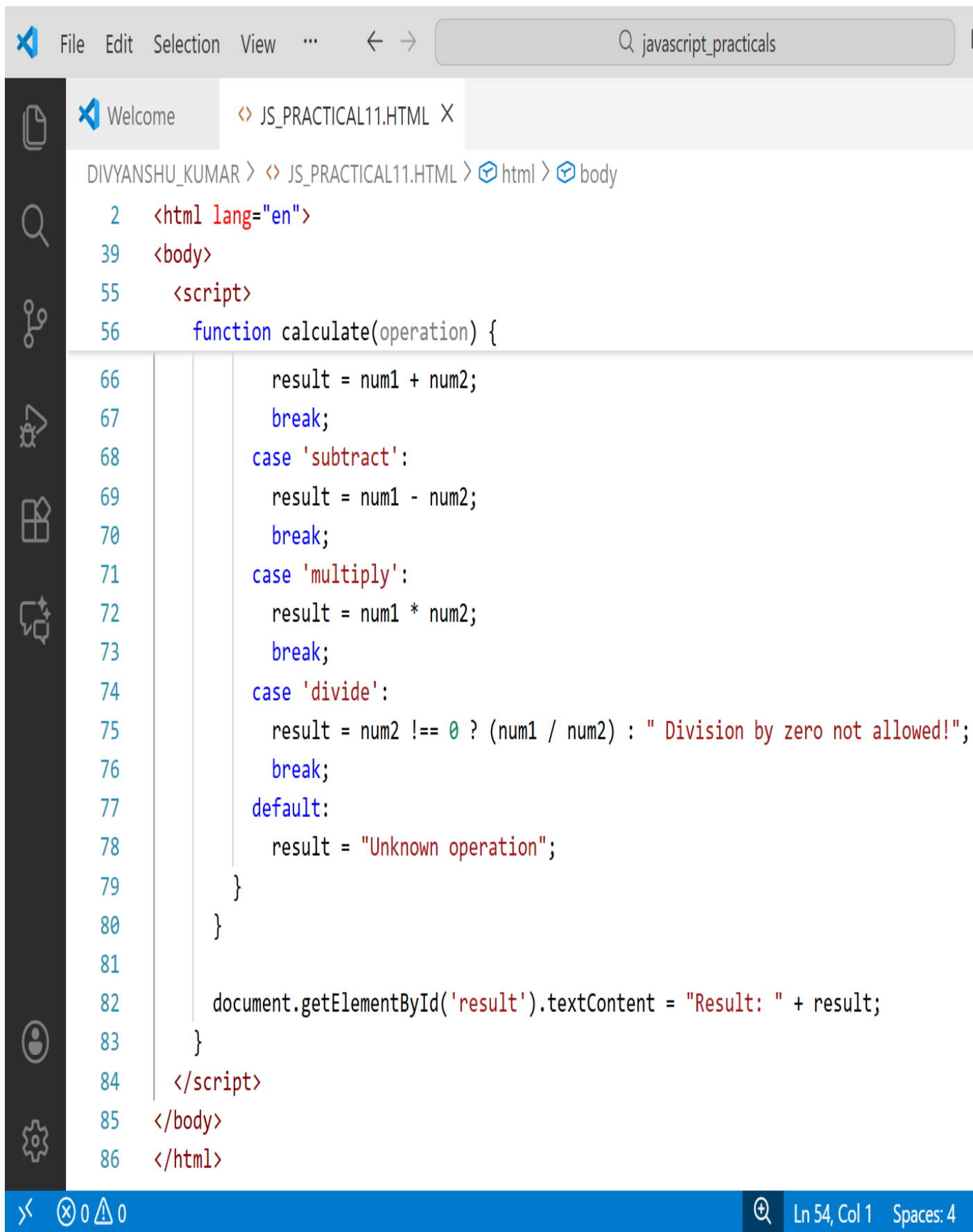
File Edit Selection View ... < > javascript_practicals

Welcome <> JS_PRACTICAL11.HTML X

DIVYANSHU_KUMAR > <> JS_PRACTICAL11.HTML > html > body

```
2 <html lang="en">
39 <body>
40 <div class="calculator">
45 <div>
46 <button onclick="calculate('add')">Add (+)</button>
47 <button onclick="calculate('subtract')">Subtract (-)</button>
48 <button onclick="calculate('multiply')">Multiply (x)</button>
49 <button onclick="calculate('divide')">Divide (÷)</button>
50 </div>
51
52 <p id="result">Result will appear here...</p>
53 </div>
54
55 <script>
56 function calculate(operation) {
57   const num1 = parseFloat(document.getElementById('num1').value);
58   const num2 = parseFloat(document.getElementById('num2').value);
59   let result;
60
61   if (isNaN(num1) || isNaN(num2)) {
62     result = " Please enter valid numbers!";
63   } else {
64     switch(operation) {
65       case 'add':
66         result = num1 + num2;
```

< 0 0 Ln 54,



File Edit Selection View ... javascript_practicals

Welcome JS_PRACTICAL11.HTML X

DIVYANSHU_KUMAR > JS_PRACTICAL11.HTML > html > body

```
2 <html lang="en">
39 <body>
55 <script>
56     function calculate(operation) {
66         result = num1 + num2;
67         break;
68     case 'subtract':
69         result = num1 - num2;
70         break;
71     case 'multiply':
72         result = num1 * num2;
73         break;
74     case 'divide':
75         result = num2 !== 0 ? (num1 / num2) : " Division by zero not allowed!";
76         break;
77     default:
78         result = "Unknown operation";
79     }
80 }
81
82 document.getElementById('result').textContent = "Result: " + result;
83 }
84 </script>
85 </body>
86 </html>
```

< 0 0 Ln 54, Col 1 Spaces: 4

OUTPUT:

Basic Arithmetic Calculator

4444

6666

Add (+)

Subtract (-)

Multiply (×)

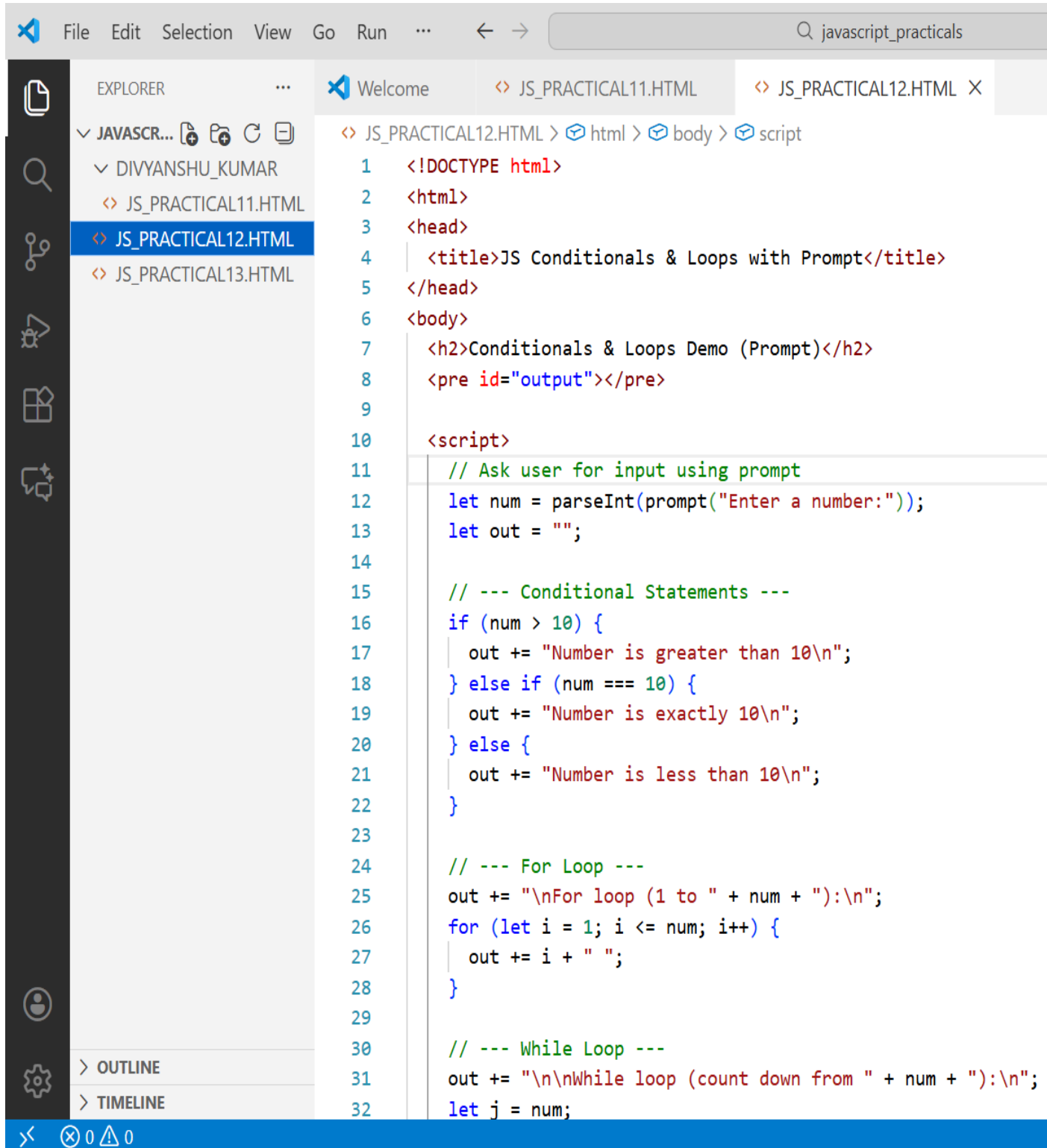
Divide (÷)

Result: 11110

EXPERIMENT -12

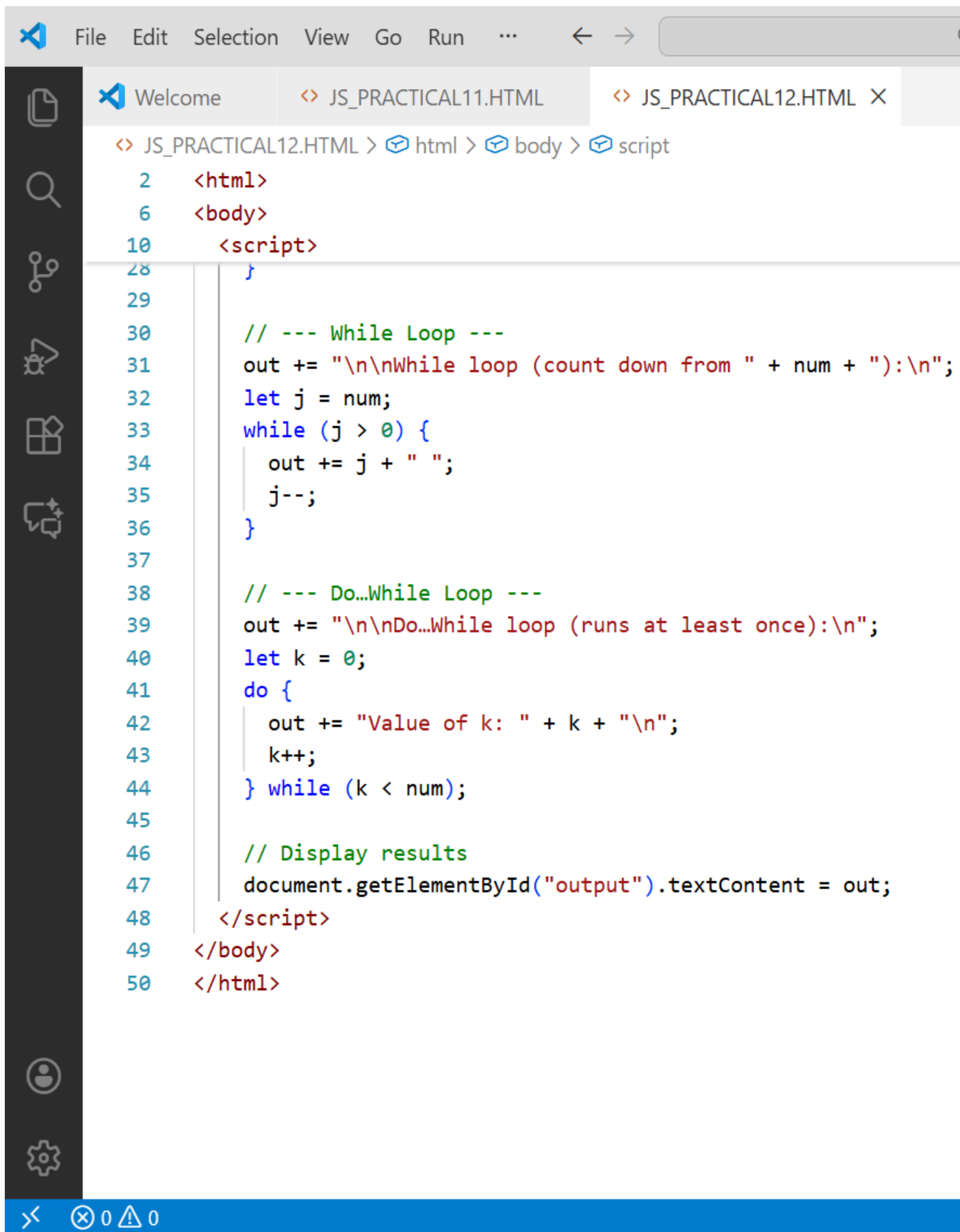
Problem statement: Write JavaScript to demonstrate conditional statements and looping (for, while, do...while)

HTML+CSS+JS CODE:



The screenshot shows the Visual Studio Code editor interface. The Explorer sidebar on the left displays a file tree with 'JAVASCRIPT_PRACTICALS' expanded, showing 'DIVYANSHU_KUMAR' and three HTML files: 'JS_PRACTICAL11.HTML', 'JS_PRACTICAL12.HTML' (selected), and 'JS_PRACTICAL13.HTML'. The main editor area shows the content of 'JS_PRACTICAL12.HTML'. The breadcrumb navigation at the top indicates the path: 'JS_PRACTICAL12.HTML > html > body > script'. The code is as follows:

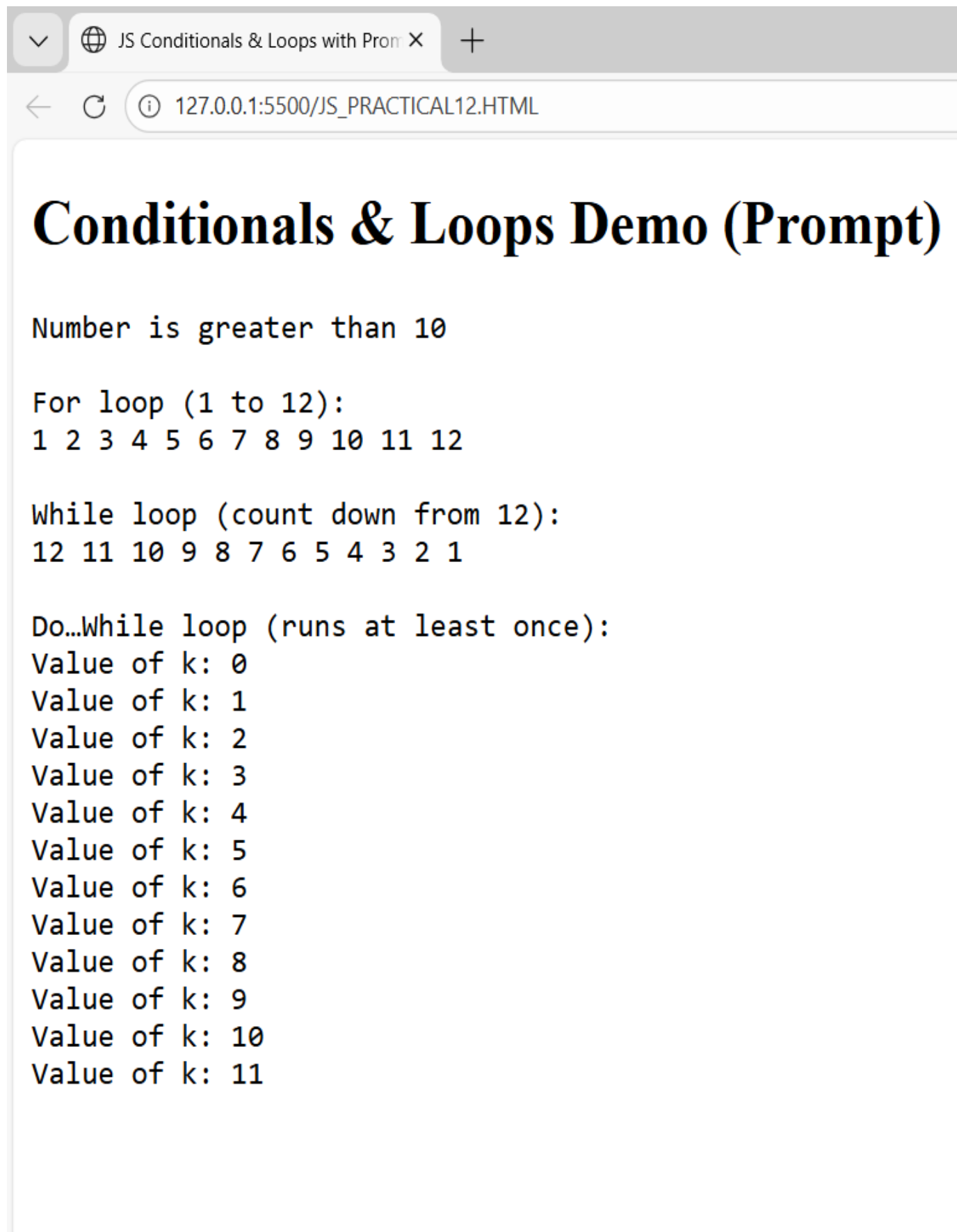
```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>JS Conditionals & Loops with Prompt</title>
5  </head>
6  <body>
7  |   <h2>Conditionals & Loops Demo (Prompt)</h2>
8  |   <pre id="output"></pre>
9
10 |   <script>
11 |       // Ask user for input using prompt
12 |       let num = parseInt(prompt("Enter a number:"));
13 |       let out = "";
14
15 |       // --- Conditional Statements ---
16 |       if (num > 10) {
17 |         out += "Number is greater than 10\n";
18 |       } else if (num === 10) {
19 |         out += "Number is exactly 10\n";
20 |       } else {
21 |         out += "Number is less than 10\n";
22 |       }
23
24 |       // --- For Loop ---
25 |       out += "\nFor loop (1 to " + num + "):\n";
26 |       for (let i = 1; i <= num; i++) {
27 |         out += i + " ";
28 |       }
29
30 |       // --- While Loop ---
31 |       out += "\n\nWhile loop (count down from " + num + "):\n";
32 |       let j = num;
```



```
File Edit Selection View Go Run ... < >
Welcome JS_PRACTICAL11.HTML JS_PRACTICAL12.HTML X
JS_PRACTICAL12.HTML > html > body > script
2 <html>
6 <body>
10 <script>
28 }
29
30 // --- While Loop ---
31 out += "\n\nWhile loop (count down from " + num + "):\n";
32 let j = num;
33 while (j > 0) {
34     out += j + " ";
35     j--;
36 }
37
38 // --- Do...While Loop ---
39 out += "\n\nDo...While loop (runs at least once):\n";
40 let k = 0;
41 do {
42     out += "Value of k: " + k + "\n";
43     k++;
44 } while (k < num);
45
46 // Display results
47 document.getElementById("output").textContent = out;
48 </script>
49 </body>
50 </html>
```

< 0 0

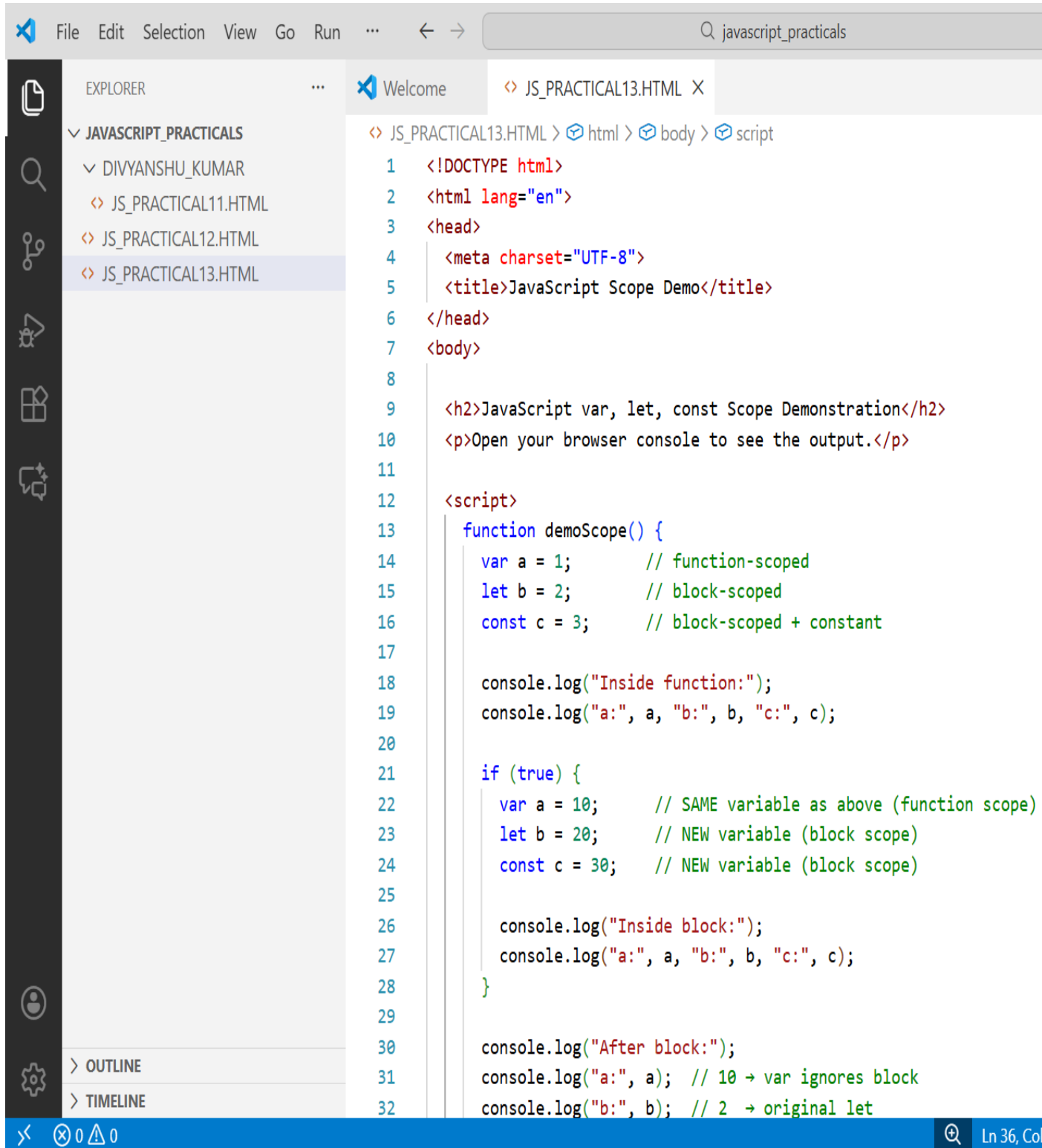
OUTPUT:



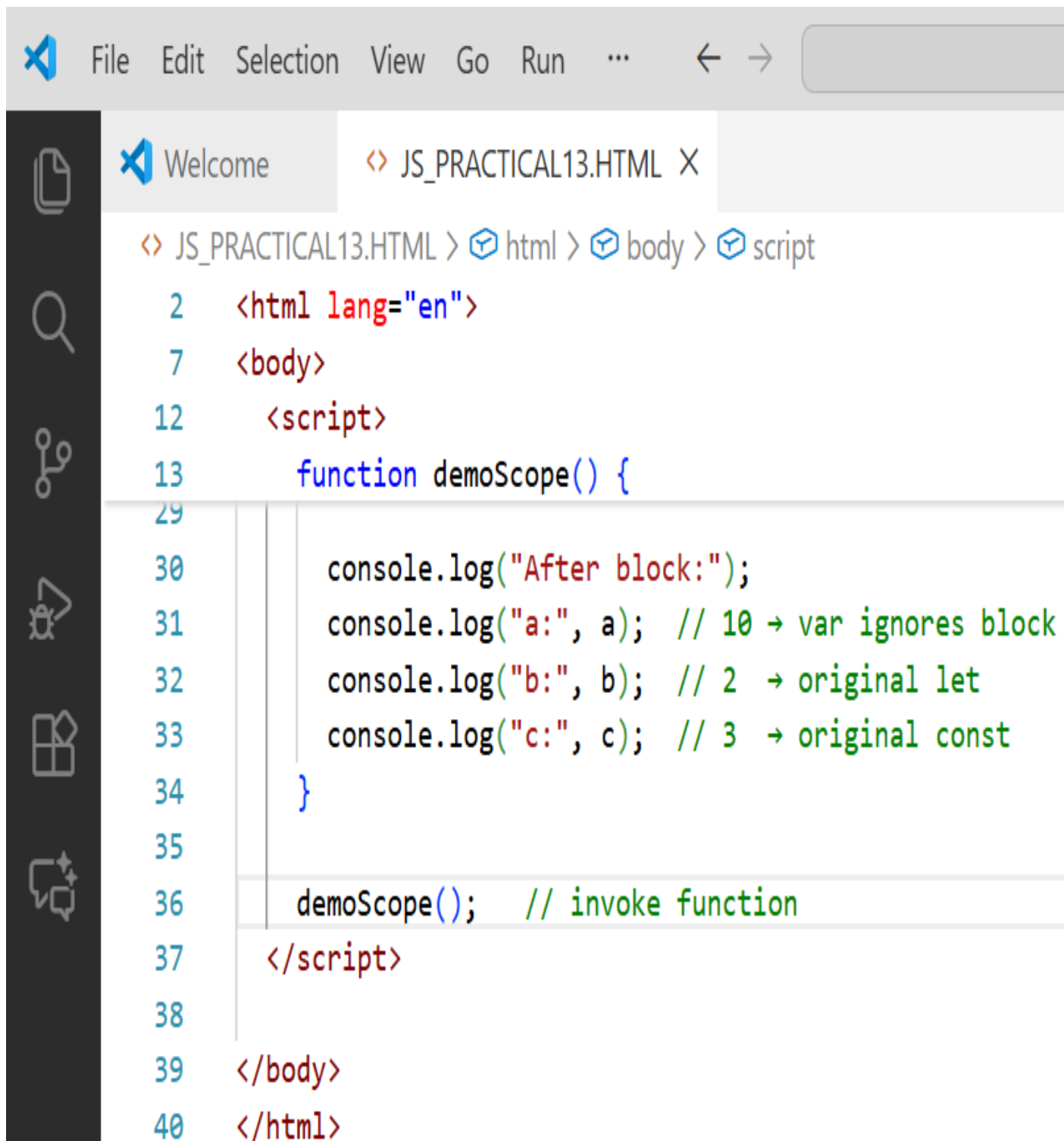
EXPERIMENT -13

Problem statement: Create and invoke user-defined functions in JavaScript; use var, let, and const for scope demonstration

HTML+CSS+JS CODE:

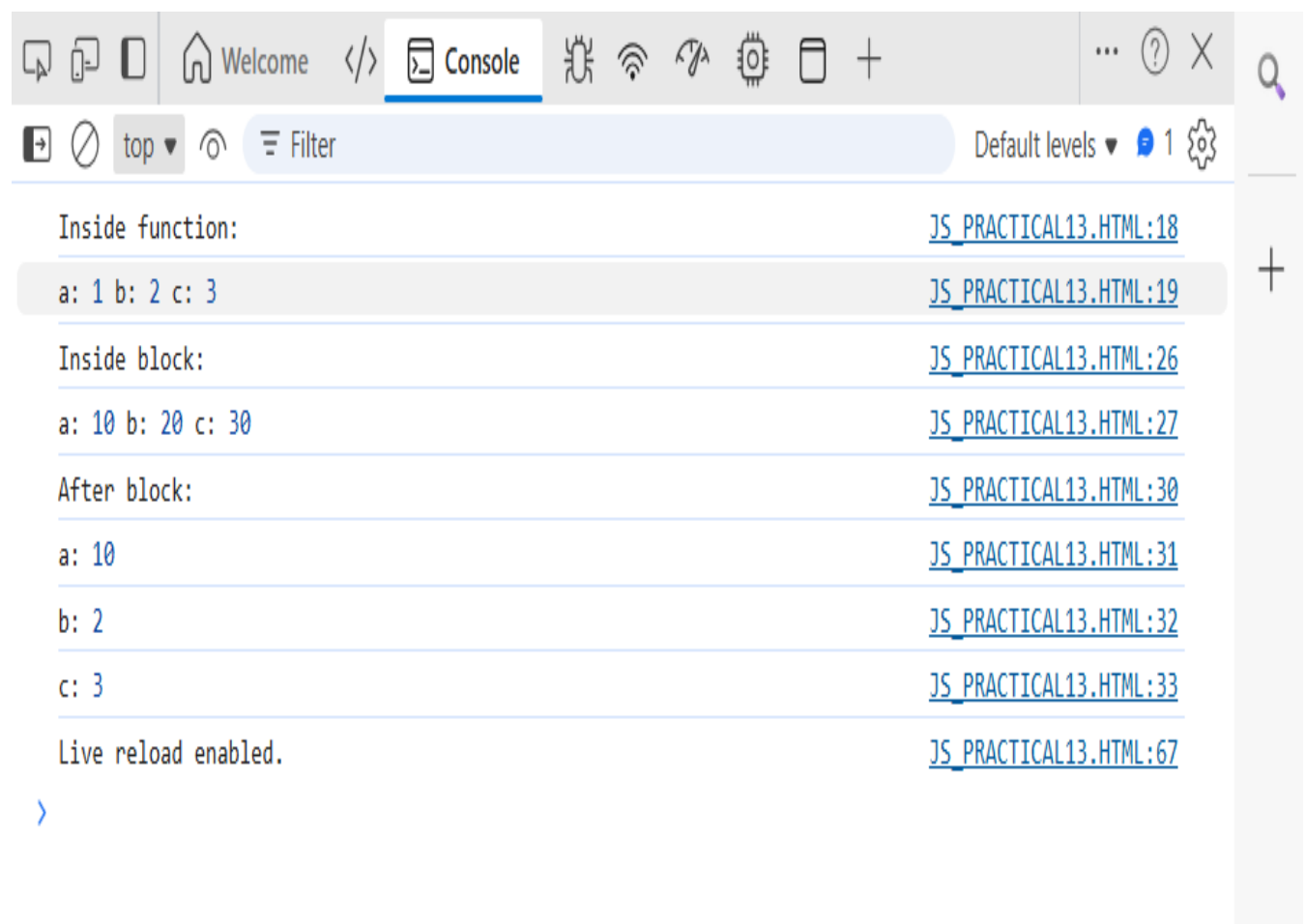
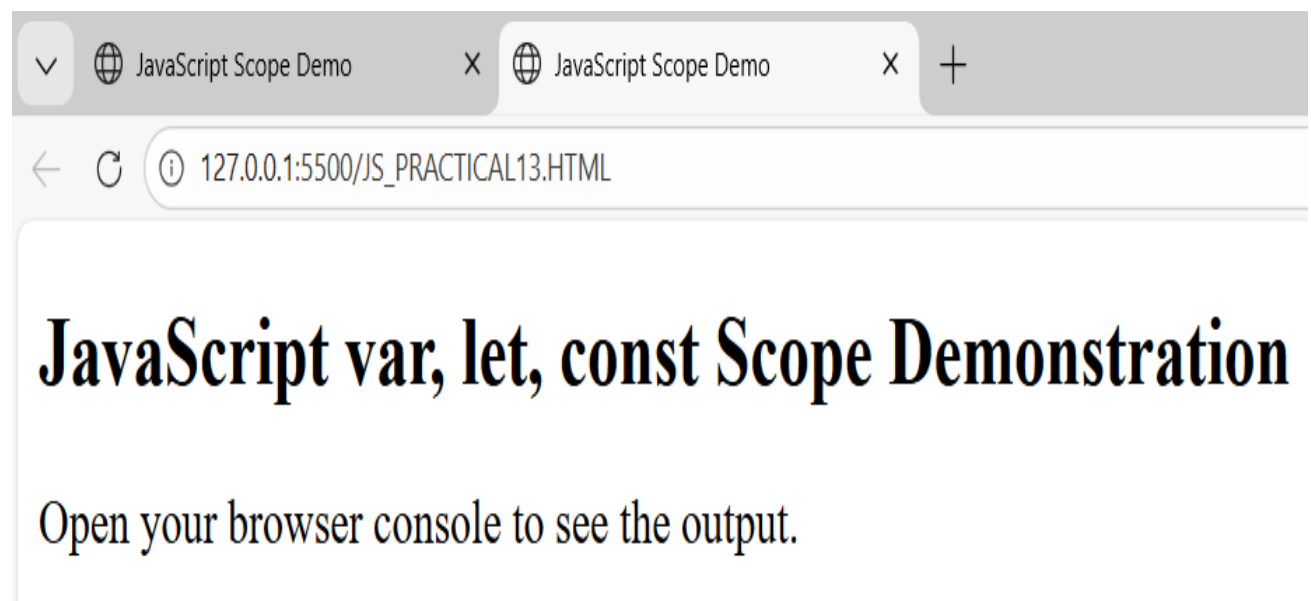


```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>JavaScript Scope Demo</title>
6 </head>
7 <body>
8
9   <h2>JavaScript var, let, const Scope Demonstration</h2>
10  <p>Open your browser console to see the output.</p>
11
12  <script>
13    function demoScope() {
14      var a = 1;      // function-scoped
15      let b = 2;      // block-scoped
16      const c = 3;    // block-scoped + constant
17
18      console.log("Inside function:");
19      console.log("a:", a, "b:", b, "c:", c);
20
21      if (true) {
22        var a = 10;   // SAME variable as above (function scope)
23        let b = 20;   // NEW variable (block scope)
24        const c = 30; // NEW variable (block scope)
25
26        console.log("Inside block:");
27        console.log("a:", a, "b:", b, "c:", c);
28      }
29
30      console.log("After block:");
31      console.log("a:", a); // 10 → var ignores block
32      console.log("b:", b); // 2 → original let
```



```
File Edit Selection View Go Run ... < >  
Welcome JS_PRACTICAL13.HTML X  
JS_PRACTICAL13.HTML > html > body > script  
2 <html lang="en">  
7 <body>  
12 <script>  
13     function demoScope() {  
29  
30         console.log("After block:");  
31         console.log("a:", a); // 10 → var ignores block  
32         console.log("b:", b); // 2 → original let  
33         console.log("c:", c); // 3 → original const  
34     }  
35  
36     demoScope(); // invoke function  
37 </script>  
38  
39 </body>  
40 </html>
```

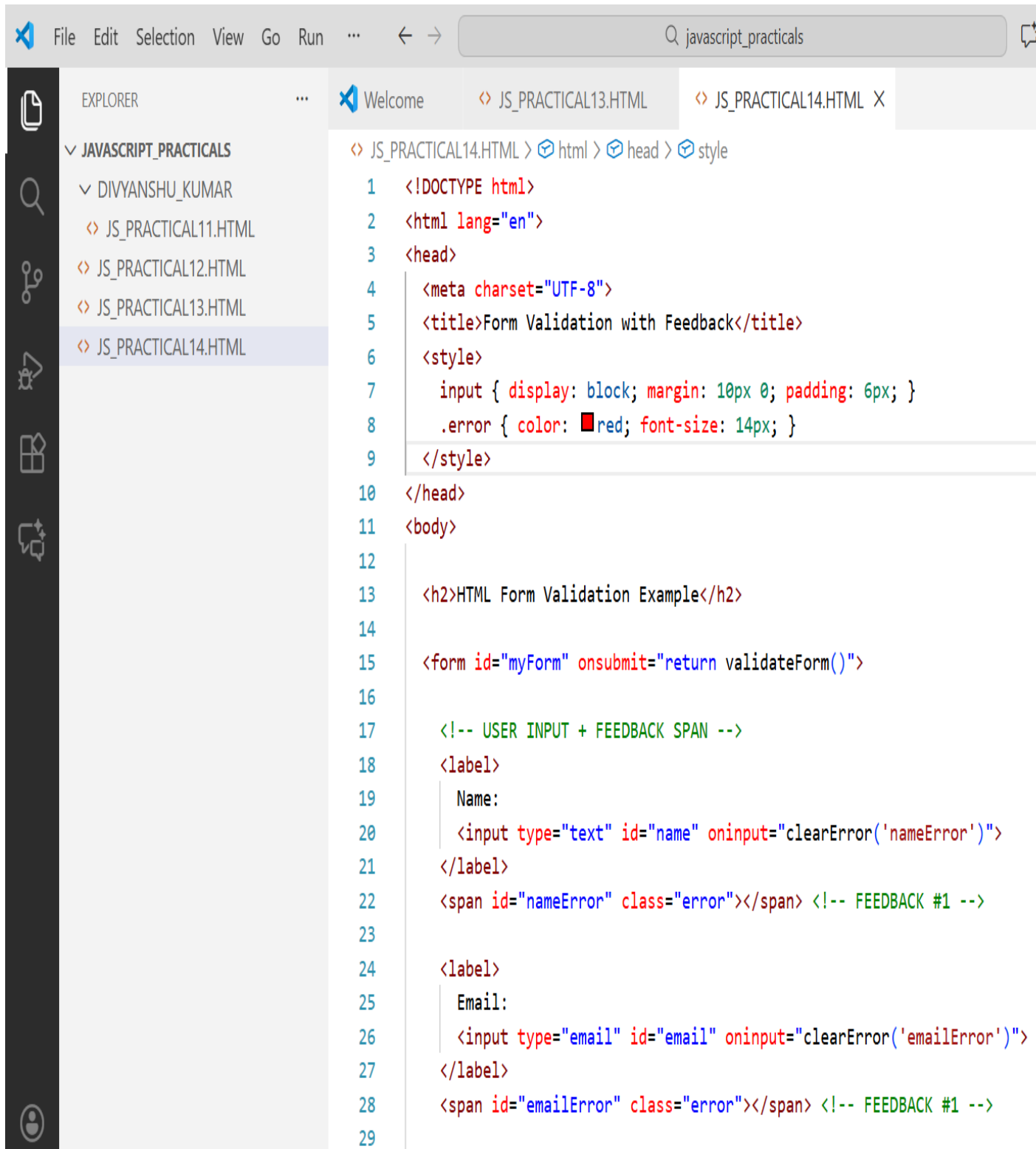
OUTPUT:



EXPERIMENT -14

Problem statement: Handle HTML form validation using JavaScript events like onsubmit, oninput, and use alert() for feedback

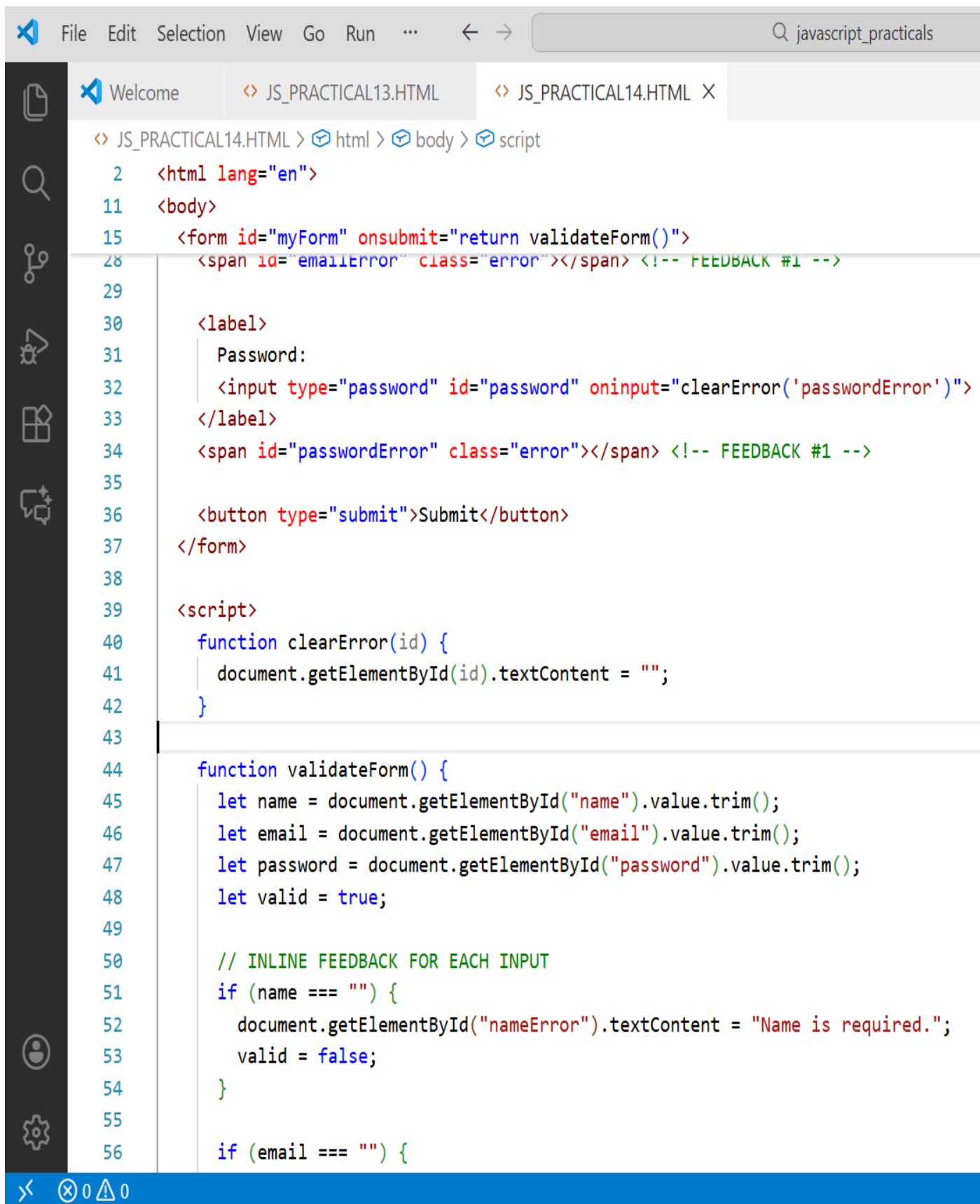
HTML+CSS+JS CODE:



The screenshot shows the Visual Studio Code editor with the following structure:

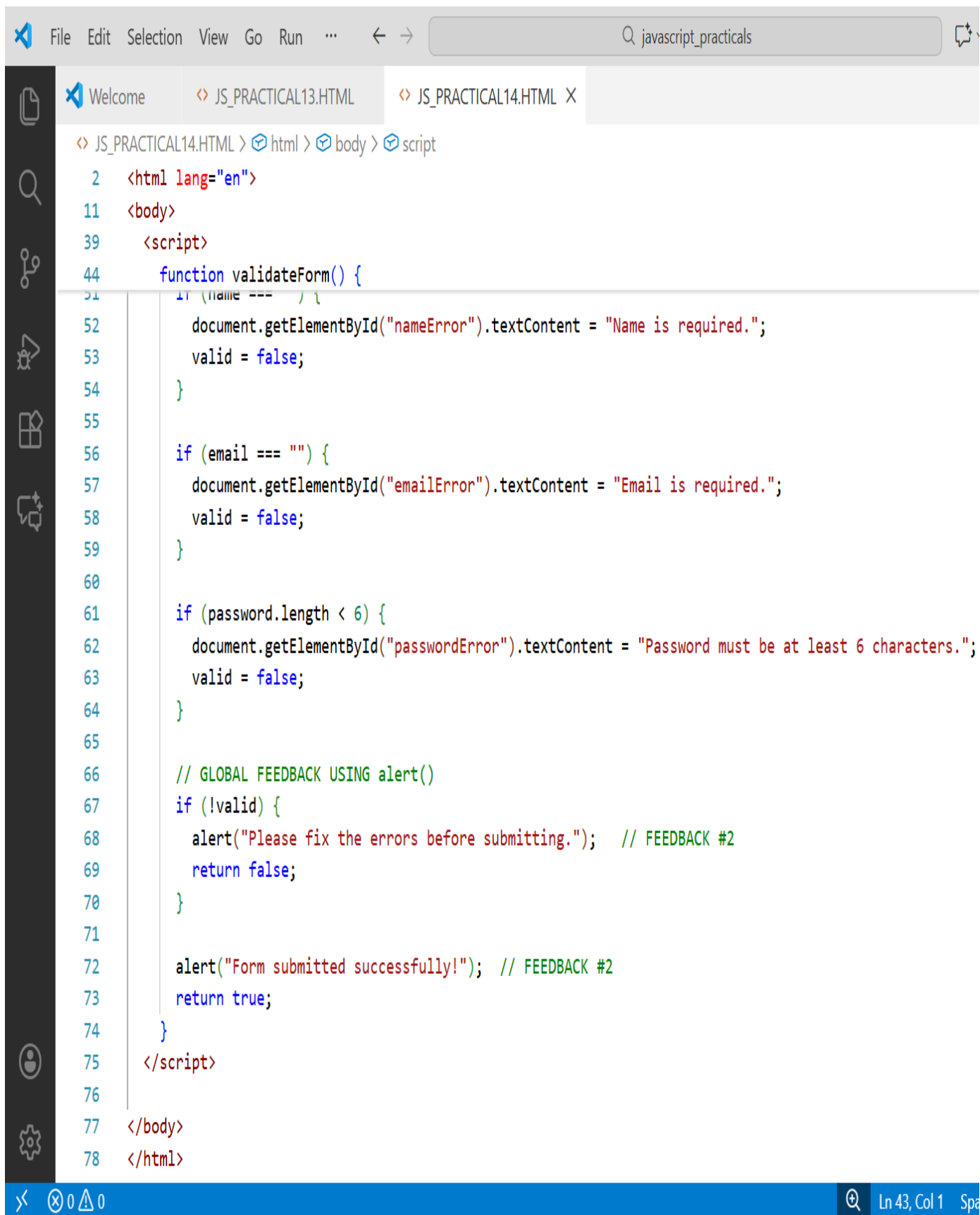
- EXPLORER:** A folder named `JAVASCRIPT_PRACTICALS` containing files `DIVYANSHU_KUMAR`, `JS_PRACTICAL11.HTML`, `JS_PRACTICAL12.HTML`, `JS_PRACTICAL13.HTML`, and `JS_PRACTICAL14.HTML`. The last file is selected.
- JS_PRACTICAL14.HTML:** Contains the following code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>Form Validation with Feedback</title>
6   <style>
7     input { display: block; margin: 10px 0; padding: 6px; }
8     .error { color: red; font-size: 14px; }
9   </style>
10 </head>
11 <body>
12
13   <h2>HTML Form Validation Example</h2>
14
15   <form id="myForm" onsubmit="return validateForm()">
16
17     <!-- USER INPUT + FEEDBACK SPAN -->
18     <label>
19       Name:
20       <input type="text" id="name" oninput="clearError('nameError')">
21     </label>
22     <span id="nameError" class="error"></span> <!-- FEEDBACK #1 -->
23
24     <label>
25       Email:
26       <input type="email" id="email" oninput="clearError('emailError')">
27     </label>
28     <span id="emailError" class="error"></span> <!-- FEEDBACK #1 -->
29
```



The screenshot shows a code editor with a sidebar on the left containing icons for Explorer, Search, Source Control, Run and Debug, Extensions, and Settings. The top menu bar includes File, Edit, Selection, View, Go, Run, and a search bar with the text 'javascript_practicals'. The editor has three tabs: 'Welcome', 'JS_PRACTICAL13.HTML', and 'JS_PRACTICAL14.HTML X'. The active tab is 'JS_PRACTICAL14.HTML', which shows the following code:

```
< JS_PRACTICAL14.HTML > html > body > script
2  <html lang="en">
11 <body>
15   <form id="myForm" onsubmit="return validateForm()">
28     <span id="emailError" class="error"></span> <!-- FEEDBACK #1 -->
29
30     <label>
31       Password:
32       <input type="password" id="password" oninput="clearError('passwordError')">
33     </label>
34     <span id="passwordError" class="error"></span> <!-- FEEDBACK #1 -->
35
36     <button type="submit">Submit</button>
37   </form>
38
39   <script>
40     function clearError(id) {
41       document.getElementById(id).textContent = "";
42     }
43
44     function validateForm() {
45       let name = document.getElementById("name").value.trim();
46       let email = document.getElementById("email").value.trim();
47       let password = document.getElementById("password").value.trim();
48       let valid = true;
49
50       // INLINE FEEDBACK FOR EACH INPUT
51       if (name === "") {
52         document.getElementById("nameError").textContent = "Name is required.";
53         valid = false;
54       }
55
56       if (email === "") {
```

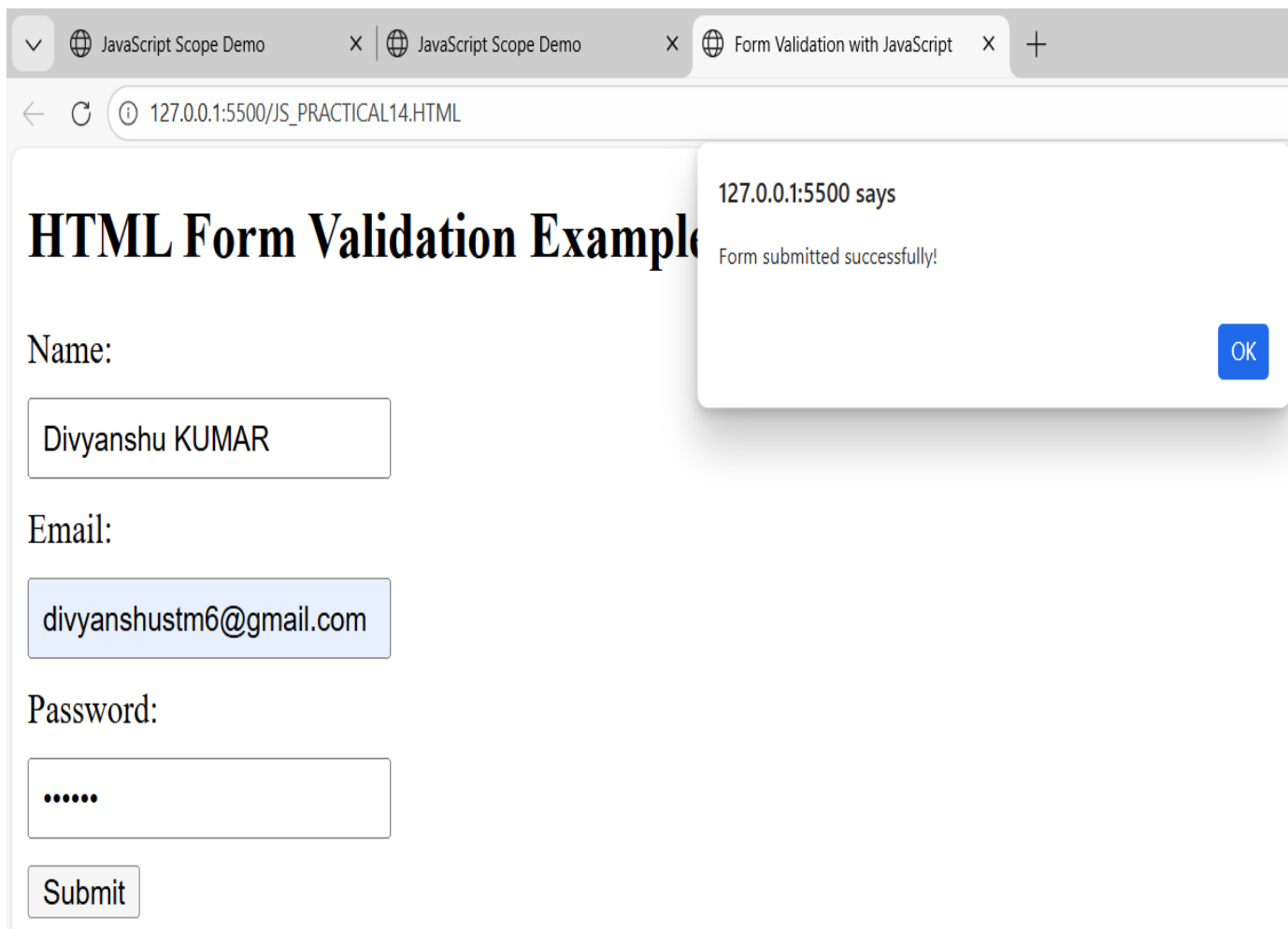


The screenshot shows a code editor with a menu bar (File, Edit, Selection, View, Go, Run, ...), a search bar containing 'javascript_practicals', and a tab bar with 'Welcome', 'JS_PRACTICAL13.HTML', and 'JS_PRACTICAL14.HTML X'. The active file is 'JS_PRACTICAL14.HTML', and the breadcrumb path is 'html > body > script'. The code is as follows:

```
2 <html lang="en">
11 <body>
39 <script>
44     function validateForm() {
51         if (name === "") {
52             document.getElementById("nameError").textContent = "Name is required.";
53             valid = false;
54         }
55
56         if (email === "") {
57             document.getElementById("emailError").textContent = "Email is required.";
58             valid = false;
59         }
60
61         if (password.length < 6) {
62             document.getElementById("passwordError").textContent = "Password must be at least 6 characters.";
63             valid = false;
64         }
65
66         // GLOBAL FEEDBACK USING alert()
67         if (!valid) {
68             alert("Please fix the errors before submitting."); // FEEDBACK #2
69             return false;
70         }
71
72         alert("Form submitted successfully!"); // FEEDBACK #2
73         return true;
74     }
75 </script>
76
77 </body>
78 </html>
```

The status bar at the bottom shows 'Ln 43, Col 1' and 'Spa'.

OUTPUT:



The screenshot shows a web browser with three tabs: 'JavaScript Scope Demo', 'JavaScript Scope Demo', and 'Form Validation with JavaScript'. The address bar shows '127.0.0.1:5500/JS_PRACTICAL14.HTML'. The main content area displays the title 'HTML Form Validation Example' in a large, bold, black serif font. Below the title, there are three form fields: 'Name:' with the value 'Divyanshu KUMAR', 'Email:' with the value 'divyanshustm6@gmail.com', and 'Password:' with masked characters '.....'. A 'Submit' button is located below the password field. A toast notification is displayed on the right side of the form, showing the text '127.0.0.1:5500 says' and 'Form submitted successfully!' with an 'OK' button.

JavaScript Scope Demo x JavaScript Scope Demo x Form Validation with JavaScript x +

127.0.0.1:5500/JS_PRACTICAL14.HTML

HTML Form Validation Example

Name:

Divyanshu KUMAR

Email:

divyanshustm6@gmail.com

Password:

.....

Submit

127.0.0.1:5500 says
Form submitted successfully!

OK